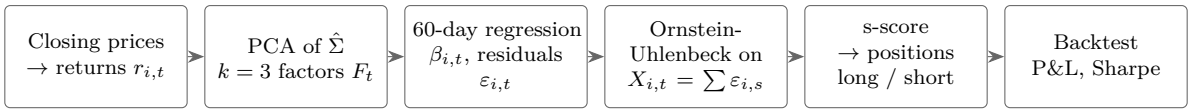
Statistical arbitrage with PCA factors

25 US stocks, 2010–2018

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Each stock's return is decomposed into a part explained by common factors, extracted by principal component analysis, and a residual specific to the stock. We then bet on the mean reversion of this residual, while hedging the factor exposure (Avellaneda & Lee, 2010). Result: the mean reversion is statistically clear, but too weak to survive transaction costs on this universe.



1. Model

For stock i and day t,

$$r_{i,t} = \alpha_i + \beta_i^\top F_t + \varepsilon_{i,t}, \qquad F_t \in \mathbb{R}^k.$$

Buying the stock and selling β_{ij} units of each factor j gives a hedged portfolio, insensitive to the market and to the sectors, with return $\alpha_i + \varepsilon_{i,t}$. This is the portfolio we trade.

2. Factors: PCA and random matrix theory

With Z the standardized returns ($T = 2264$ days, $N = 25$ stocks), we diagonalize the correlation matrix:

$$\hat{\Sigma} = \frac{Z^\top Z}{T-1} = V \Lambda V^\top.$$

The 25 stocks are split into five sectors of five (banks, energy, technology, healthcare, consumer staples): the sector structure is directly visible in $\hat{\Sigma}$.

For a pure-noise matrix, the eigenvalues stay within the Marchenko-Pastur interval:

$$\lambda_\pm = \left(1 \pm \sqrt{N/T}\right)^2, \qquad \lambda_+ = 1.22.$$

Three eigenvalues lie above it (11.1, 2.2, 1.5): the market and two sector oppositions. We keep $k = 3$ eigenportfolios, with weights and returns

$$Q = D^{-1} V_k, \qquad F = R\,Q, \qquad D = \mathrm{diag}(\sigma_1, \ldots, \sigma_N).$$

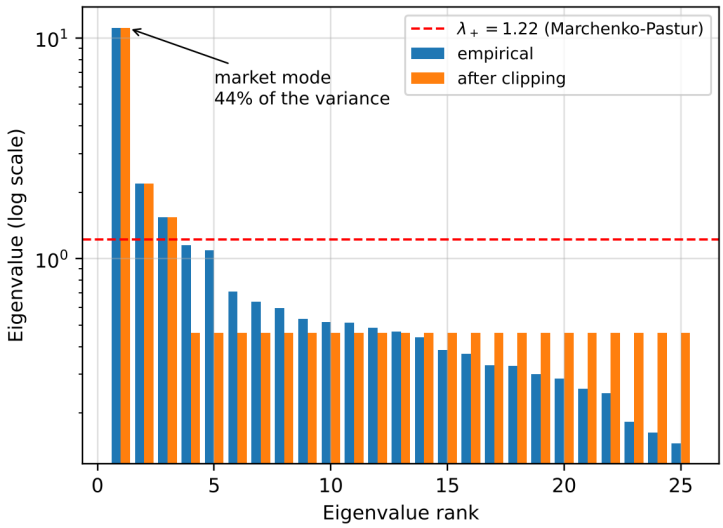


Figure 1: Eigenvalues of $\hat{\Sigma}$ (log scale). Only three exceed the noise threshold λ_+; Bouchaud-Potters clipping replaces all the others by their mean.

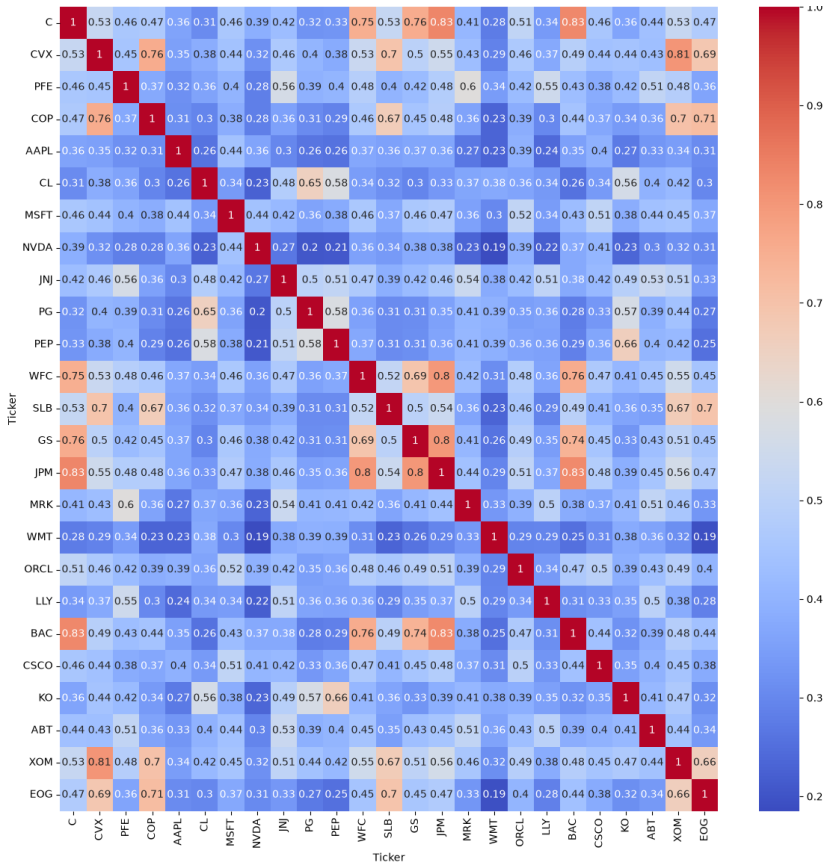


Figure 2: Correlation matrix of daily returns, 2010–2018. All correlations are positive (market mode); the warmer blocks correspond to the sectors, in particular banks (JPM, BAC, C, WFC, GS) and energy (XOM, CVX, COP, EOG, SLB).

Two denoising methods are compared: Bouchaud-Potters clipping (eigenvalues below λ_+ replaced by their mean, which preserves the trace) and Ledoit-Wolf shrinkage towards a constant-correlation matrix C,

$$\hat{\Sigma}_{LW} = (1-\delta)\,\hat{\Sigma} + \delta\, C, \qquad C_{ij} = \bar{\rho} \;\; (i \neq j),$$

with δ estimated at 0.039. With $N/T \approx 0.01$, the estimation noise is small: shrinkage barely moves the spectrum and leaves the eigenvectors unchanged.

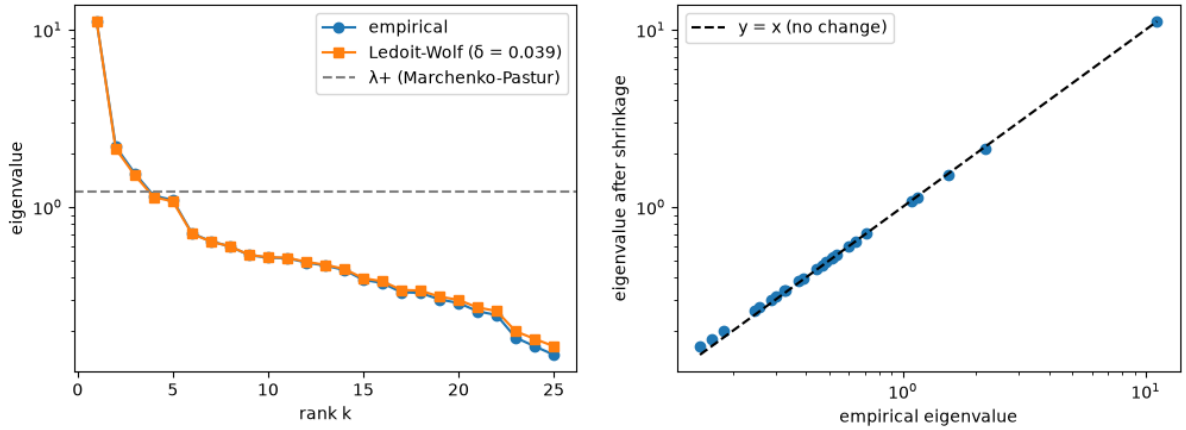


Figure 3: Effect of Ledoit-Wolf shrinkage ($\delta = 0.039$): spectra before and after (left), and eigenvalues after shrinkage against before (right), almost all on the diagonal.

3. Residuals and mean reversion

The $\beta_{i,t}$ are estimated by least squares over the 60 days preceding t, with a one-day lag so that no future information is used. The cumulative residual $X_{i,t} = \sum_{s \leq t} \varepsilon_{i,s}$ is the performance of the hedged portfolio.

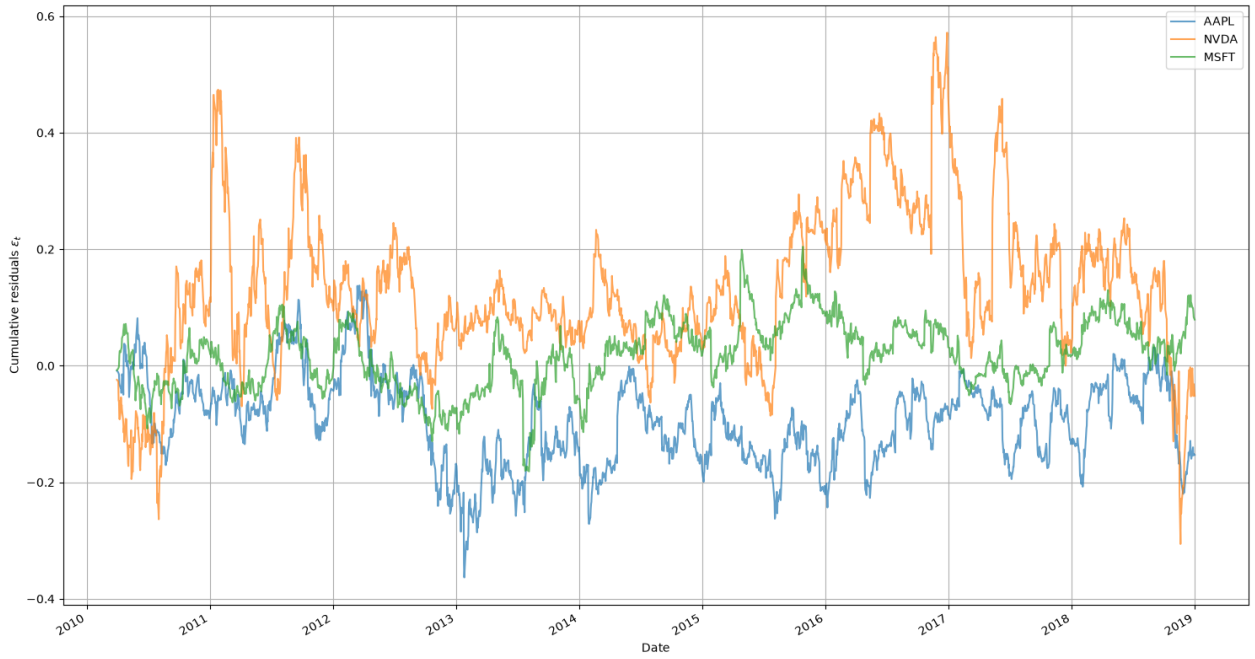


Figure 4: Cumulative residuals of three technology stocks. They oscillate around an equilibrium level, with no common trend; NVDA shows the widest swings.

The augmented Dickey-Fuller test (H_0: random walk) is rejected at the 5% level for **21 stocks out of 25**, and at the 1% level for 17. Three of the four exceptions are oil stocks (XOM, EOG, SLB): the oil price is probably not fully captured by three factors.

4. Signal: Ornstein-Uhlenbeck process

We model $X_{i,t} = \sum_{s \le t} \varepsilon_{i,s}$ as

$$dX_t = \kappa\,(m - X_t)\,dt + \sigma\,dW_t,$$

whose exact discretization with a one-day step is an AR(1), estimated on each 60-day window:

$$X_{n+1} = a + b\,X_n + \zeta_n, \qquad b = e^{-\kappa}, \quad m = \frac{a}{1-b}, \quad \sigma_{eq} = \sqrt{\frac{\mathrm{Var}(\zeta)}{1-b^2}}.$$

A stock is only traded if $0 < b < 1$ and its half-life $\ln 2/\kappa$ is below 30 days. The signal is the normalized distance to equilibrium,

$$s_{i,t} = \frac{X_{i,t} - m_i}{\sigma_{eq,i}}, \qquad \mathbb{E}\left[X_{t+h} - X_t\right] = -\,s_t\,\sigma_{eq}\left(1 - e^{-\kappa h}\right),$$

so the expected gain is proportional to $-s_t$. Empirically, the s-scores have a mean close to 0 and a standard deviation between 0.87 and 0.99 depending on the stock.

5. Strategy and backtest

Avellaneda and Lee rules, with a position $q_{i,t} \in \{-1, 0, +1\}$ on the hedged portfolio:

$$q_{i,t} = \begin{cases} +1 & \text{if } q_{i,t-1} = 0 \text{ and } s_{i,t} < -1.25 \\ -1 & \text{if } q_{i,t-1} = 0 \text{ and } s_{i,t} > 1.25 \\ 0 & \text{if } q_{i,t-1} = -1 \text{ and } s_{i,t} < 0.75, \text{ or } q_{i,t-1} = +1 \text{ and } s_{i,t} > -0.5 \\ q_{i,t-1} & \text{otherwise} \end{cases}$$

Each position uses $1/N$ of the capital and earns the return of the hedged portfolio the next day:

$$\mathrm{P\&L}_t = \frac{1}{N} \sum_{i=1}^{N} q_{i,t-1}\,(\alpha_i + \varepsilon_{i,t})\,.$$



Figure 5: Portfolio value for €1 invested, with $1/N$ of the capital per position, without fees and with fees of 5 basis points.

	no fees	5 bp fees
Annual return	0.63%	−0.14%
Annual volatility	1.74%	1.73%
Sharpe ratio	0.36	−0.08
Max drawdown	−4.2%	−6.6%
Winning days	49.4%	48.1%
Correlation with market	0.00	0.00

Table 1: Backtest metrics, 2010–2018. Returns in excess of the risk-free rate.

6. Conclusion and limitations

The strategy is indeed market-neutral, but its gross Sharpe ratio of 0.36, earned mostly in 2016–2018, disappears with fees of 5 basis points. The mean reversion of the residuals is real in the statistical sense, but too weak to be exploited robustly on 25 large caps between 2010 and 2018, which matches the erosion of these strategies observed by Avellaneda and Lee after the early 2000s.

Limitations. PCA estimated over the whole period (look-ahead bias, to be removed with a rolling PCA); short-selling costs not modelled; thresholds taken from the paper without optimization; small universe chosen ex post (survivorship bias).

Next steps. $k = 4$ or 5 for energy, β estimated by Kalman filter, risk-based position sizing, a universe of several hundred stocks.

Python (numpy, pandas, statsmodels). Reference: Avellaneda & Lee, *Statistical arbitrage in the U.S. equities market*, Quantitative Finance, 2010.